

Statistical Foundations for Credible Human–AI Teaming Claims

TEAM-Audit: A Two-Endpoint Audit of Margins

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Abstract

Human–AI decision systems are often called “teams” when artificial intelligence (AI) assistance improves human performance. That comparison cannot carry the label: a system can help the people using it and still be worse than the AI working alone. We propose TEAM-Audit, a two-baseline audit for the performance version of the claim: evaluate the final, frozen system on data not used to build it; compare it separately with human-only and with AI-only performance; and only when both comparisons are supported with uncertainty call the result performance complementarity — “beats both.” For systems that route cases between a human and an AI, the audit corrects historical logs that record only the acting party’s outcome, and it reports human workload beside accuracy. In simulation, ignoring development-stage uncertainty produced false “beats both” verdicts in 0.504 of null replications, and reusing one sample to both select and evaluate a policy produced 0.103. Reanalyzing a public human–AI study, all 14 assisted conditions beat both baselines when the study’s questions are treated as fixed, but only 2 of 14 when the uncertainty also covers the broader item population those questions stand in for. The verdict turns on a choice that is almost never stated.

Keywords

human–AI teaming, complementarity, evaluation methodology, intersection–union tests, off-policy evaluation

1 Introduction

Suppose unaided people decide correctly 75% of the time, an artificial intelligence (AI) model alone reaches 90%, and people working with the model reach 82%. The assistance clearly helped. Yet the combined system is still eight points worse than the model alone, and calling the pair a successful team overstates the evidence. Throughout this paper, **performance complementarity** means that the human–AI system performs better than the human alone and better than the AI alone, on the same outcome and population of cases; for short, it beats both. This is a narrow performance claim: it does not say the human and the AI share goals, coordinate, adapt, or trust each other, only that the combination outperformed both of its parts.

Arrangements of this shape now sit inside decisions in medicine, content moderation, credit, and law, and they are routinely described as human–AI teams. The description carries weight: a “better together” claim can justify keeping people in a costly loop, support deployment decisions, and propagate through reviews and meta-analyses. Yet surveys find that combined systems are usually benchmarked against unaided humans rather than against the

model alone [Lai et al. 2023; Vaccaro et al. 2024], so “the AI helped” is reported where “the pair beat the AI” was never tested.

A large literature already builds these systems well. Learning-to-defer methods decide who handles each case [Madras et al. 2018; Mozannar et al. 2023; Mozannar and Sontag 2020; Verma and Nalisnick 2022]; expert reliability models decide when a human’s judgment can be trusted [Raghu et al. 2019; Steyvers et al. 2022]; complementary-policy learning allocates work [Gao et al. 2023; Wilder et al. 2020]; empirical studies measure whether assistance helps [Bansal et al. 2021; Hemmer et al. 2025; Vaccaro et al. 2024; Xu et al. 2026]. Bansal et al. [2021] and others already define complementary performance as exceeding both solo alternatives; the criterion is not ours.

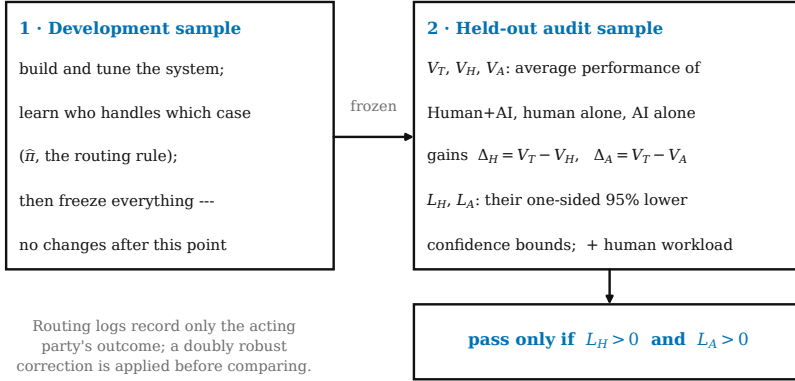
What none of this settles is the evidentiary question about a finished system: do we have reliable evidence that it beats both alternatives? Five common evaluation choices make the answer unreliable. Many studies compare only with humans. A routing rule is often selected and evaluated on the same data, so the reported gain includes selection optimism. Historical logs are selective — only the acting party’s outcome is recorded — so raw human and AI averages describe different mixes of cases. The human labor a system consumes is rarely reported next to its accuracy. And the population the claim generalizes over is usually left implicit. These are evaluation problems rather than optimization problems, and Table 1 summarizes them.

Table 1: Common evaluation choices, why each misleads, and the audit’s response.

Common choice	Why it can mislead	Audit response
Compare only with humans	The AI alone may still be better	Compare with both baselines
Tune and test on the same data	Selection optimism inflates the gain	Freeze, then audit held out
Average raw routing logs	Humans and AI saw different cases	Correct for case assignment
Report accuracy alone	The gain may cost human labor	Report workload too
Leave the population implicit	The uncertainty answers a different question	State what the claim covers

This paper proposes TEAM-Audit (Two-Endpoint Audit of Margins), a two-baseline audit of the finished system; the two endpoints are the margins over the two baselines. The audit asks two questions: is the combined system better than the human alone, and is it better than the AI alone? It answers both on data the system has never seen, with one-sided confidence bounds, and claims

(a) The TEAM-Audit protocol



(b) The four possible outcomes

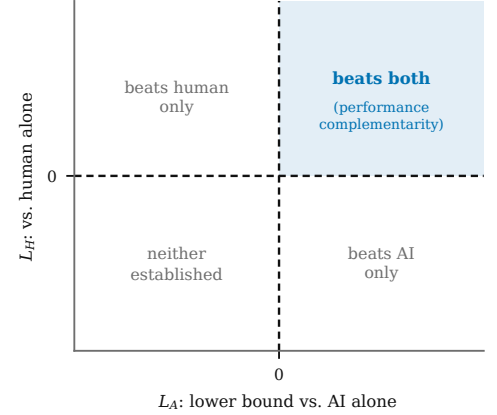


Figure 1: The TEAM-Audit protocol and its outcomes. (a) Build and freeze on development data; a held-out audit sample yields the three performance levels V_T, V_H, V_A (Human+AI, human alone, AI alone), the gains $\Delta_H = V_T - V_H, \Delta_A = V_T - V_A$, and their one-sided 95% lower bounds L_H, L_A , with human workload beside them; routing logs first receive a doubly robust correction. (b) The four outcomes as regions of the (L_A, L_H) plane (labels of Table 2); no data shown.

performance complementarity only when both bounds clear zero. In assisted interaction, where a person decides with AI advice on screen, a randomized study estimates all three performance levels directly; in selective delegation, where a frozen rule routes each case to the human or the AI, historical logs need a statistical correction, because only the chosen actor’s outcome was recorded. The audit also reports the human workload the system consumed.

Both empirical parts measure what the safeguards prevent. In simulation, an evaluator that ignores development-stage uncertainty declared false “beats both” verdicts in 0.504 of null replications, and reusing one sample to search and to evaluate raised the false rate to 0.103. The held-out audit held both at or below the nominal 0.05. Reanalyzing Bansal et al. [2021], every one of 14 assisted conditions beats both baselines in point estimate, and the audited verdict then turns entirely on which cases the claim is about. All 14 conditions beat both for the study’s own questions, 2 for the broader item population those questions stand in for.

Contributions. (i) An evaluation standard: two required comparisons, stated as an explicit statistical hypothesis. (ii) TEAM-Audit, a protocol for learned routing systems: freeze, then evaluate held out, with a selective-log correction, a development workload budget that never overshoots, and workload reporting. (iii) Evidence: simulations separating which safeguards matter, and a reanalysis showing that the choice of target population, rarely made explicit, can decide the verdict. We are not defining social teamwork. The audit tests one performance implication that routinely rides along with the word “team.”

2 Related work

The system-building methods of Section 1 (deferral, calibrated variants, complementary-policy learning) solve design problems. Our

question starts where they stop. Once a system exists, what evidence supports the claim that it beats both alternatives?

Empirical and conceptual work on complementarity supplies the criterion we formalize. Bansal et al. [2021] showed that explanations increase acceptance of AI advice without reliably producing complementary performance; Bayesian and conceptual accounts characterize when complementarity is attainable [Hemmer et al. 2025; Steyvers et al. 2022]; meta-analysis and recent cross-task evidence find that combinations frequently fail to beat the stronger solo actor [Vaccaro et al. 2024; Xu et al. 2026]; and a broad literature studies over-reliance [Bućinca et al. 2021; Green and Chen 2019; Schemmer et al. 2023]. We turn the shared criterion into an explicit hypothesis with stated uncertainty. Alur et al. [2023] also give an inferential treatment, but of whether experts carry signal no algorithm on the available features could recover, stated upstream of deployment. We test whether a fielded arrangement beat both solo alternatives. The word “team” itself belongs to human-factors and organizational research, which defines teaming through shared goals, coordination, adaptation, and calibrated trust [Groom and Nass 2007; Klein et al. 2004; National Academies of Sciences, Engineering, and Medicine 2022; O’Neill et al. 2022; Seeber et al. 2020]. TEAM-Audit brackets those relational criteria.

The estimators are standard: inverse-probability weighting (IPW), doubly robust (DR) estimation, and policy evaluation from logged data [Athey and Wager 2021; Chernozhukov et al. 2018; Dudík et al. 2011; Kitagawa and Tetenov 2018; Robins et al. 1994]. The contribution is the protocol combining them into a transparent two-baseline audit.

3 What “better together” means

This section pins the claim down in plain terms: the three quantities being compared, the four outcomes the comparison can produce, and the two arrangements the audit covers.

3.1 The three performance levels

Write V_T for the average performance of the combined human-AI intervention, V_H for the human-only alternative, and V_A for the AI-only alternative, on one pre-specified outcome and population of cases. The two gains are

$$\Delta_H = V_T - V_H, \quad \Delta_A = V_T - V_A. \quad (1)$$

The first asks whether the system beats the human alone, the second whether it beats the AI alone; a performance-complementarity claim asserts both are positive. The population matters as much as the outcome: Section 6 shows the two targets receiving different verdicts from the same data.

3.2 Four possible evidence outcomes

Table 2 names the four outcomes, and Figure 1b draws them as regions of the (L_A, L_H) plane. The labels are statements about evidence, not mechanism. “Beats human only” means the AI comparison was not resolved at this sample size, never that complementarity was ruled out.

Table 2: The four evidence outcomes. Bold marks the only outcome supporting a performance-complementarity claim; parenthetical terms connect to prior work’s vocabulary.

vs. human	vs. AI	Label
better	better	beats both (complementarity)
better	not established	beats human only (AI augmentation)
not established	better	beats AI only (human oversight)
not established	not established	neither established

3.3 Two human-AI settings

In **assisted interaction**, a person makes the final decision with AI output on screen; a randomized experiment, with the model’s predictions recorded per item, estimates all three performance levels directly. In **selective delegation**, a routing rule sends each case either to the human or to the AI, and the natural data are historical logs, which are selective: only the acting party’s outcome is recorded. If the router sent easy cases to the AI, the log’s human accuracy describes hard cases and its AI accuracy easy ones, so raw averages compare case mixes, not actors. Section 4 states the correction. Throughout, held-out data means data that played no role in building, tuning, or selecting the system.

4 How TEAM-Audit works

This section states the protocol, then the estimation in each arrangement, the workload rule, and the conditions under which the inference is valid.

4.1 The audit in five steps

- (1) **Pre-specify** the outcome, target population, significance level, and, for routing systems, one human-workload budget.
- (2) **Build** on development data: fit models, learn the routing rule, tune freely.

- (3) **Freeze** the system; nothing after this point may change it.
- (4) **Audit** on held-out data: estimate joint, human-only, and AI-only performance; form both gains with one-sided 95% lower bounds, correcting for selective logging when auditing from historical data.
- (5) **Report** both gains, both bounds, and realized human workload; claim “beats both” only if both bounds exceed zero.

Steps 2–3 make step 4 meaningful: the audit evaluates the frozen system that development actually produced.

4.2 Assisted interaction

A randomized assisted study estimates V_T and V_H from its assigned conditions and V_A from the model’s recorded predictions on the same items, so no selective-log correction is needed. What still requires care is the uncertainty. The bounds must resample whatever the claim generalizes over, participants if the claim is about new people, items as well if it is about new cases.

4.3 Learned delegation from selective logs

Each case has features X , two potential rewards $Y^H, Y^A \in [0, 1]$ — what the human, or the AI, would score on it — and a logged action $Z \in \{H, A\}$ with observed reward $Y = 1\{Z = H\}Y^H + 1\{Z = A\}Y^A$, logged with propensity $e_z(x) = \Pr(Z = z \mid X = x)$. A policy π assigns the human with probability $\pi_H(x)$. The frozen router below realizes any tie randomization through an immutable case identifier U , independent of (Y^H, Y^A) given X . Writing $A_\pi(x, u) \in \{H, A\}$ for the realized rule, $\pi_H(x) = \Pr[A_\pi(x, U) = H \mid X = x]$. The policy’s value is $V(\pi) = \mathbb{E}[\pi_H(X)\mu_H(X) + (1 - \pi_H(X))\mu_A(X)]$ for $\mu_z(x) = \mathbb{E}[Y^z \mid X = x]$. The constant policies π^H and π^A recover V_H and V_A , and $\Delta_j(\pi)$ denotes the corresponding margin.

Split the data into a development sample $\mathcal{D}_{\text{dev}}$ of size n_d and an independent audit sample $\mathcal{D}_{\text{audit}}$ of size n_a . On $\mathcal{D}_{\text{dev}}$, fit actor outcome models $\hat{\mu}_H, \hat{\mu}_A$, and take $\hat{e}_z$ to be the logging propensity, known exactly or fitted on $\mathcal{D}_{\text{dev}}$ and frozen. Form the doubly robust human-minus-AI pseudo-outcome $\Gamma_i = \hat{\mu}_H - \hat{\mu}_A + \frac{1\{Z_i=H\}}{\hat{e}_H}(Y_i - \hat{\mu}_H) - \frac{1\{Z_i=A\}}{\hat{e}_A}(Y_i - \hat{\mu}_A)$, fit a simple score $\hat{g}(x) \approx \mathbb{E}[\Gamma \mid X = x]$, positive where the human looks better, build the budget-capped router $\hat{\pi}$ from it (Section 4.4), and freeze everything. The audit then needs one estimator per case,

$$\psi_i(\pi) = \sum_{z \in \{H, A\}} \pi_z(X_i) \left[\hat{\mu}_z(X_i) + \frac{1\{Z_i=z\}}{\hat{e}_z(X_i)} \{Y_i - \hat{\mu}_z(X_i)\} \right]. \quad (2)$$

In words, ψ_i starts from the frozen model’s guess for the actor the policy would use and corrects it with the observed outcome whenever the log contains that actor’s answer, reweighted by how likely that was. Averaging $\psi_i(\pi)$ estimates $V(\pi)$; the paired differences $d_{i,H} = \psi_i(\hat{\pi}) - \psi_i(\pi^H)$ and $d_{i,A} = \psi_i(\hat{\pi}) - \psi_i(\pi^A)$ estimate the gains, with $\hat{\Delta}_j$ the mean of the $d_{i,j}$ and one-sided Wald bounds $L_j = \hat{\Delta}_j - z_{0.95}\widehat{\text{SE}}_j$, where $z_{0.95}$ is the standard-normal 0.95 quantile and the standard error (SE) $\widehat{\text{SE}}_j$ is the sample standard deviation of the $d_{i,j}$ over $\sqrt{n_a}$. Identification requires consistency, conditional exchangeability ($Y^H, Y^A \perp Z \mid X$ (no unmeasured confounders), and overlap (every case had some probability of going to each actor); valid inference additionally requires bounded rewards and audit observations independent of everything used to build the policy.

4.4 Human workload is part of the result

The confirmatory policy caps human assignment at $B_0 = 0.20$ of the development sample. Eligibility requires $\hat{g}(x) > 0$; ranking eligible cases by $\hat{g}$, the cutoff c_B is the score of the $\lfloor B_0 n_d \rfloor$ -th such case, or the smallest eligible score when fewer are eligible. A stratum model makes cases tie at c_B in bulk. With m tied and r slots unspent, we rank the m on a frozen hash $h(u)$ of an immutable case identifier and admit the r smallest; the threshold h_B is frozen with the policy, and a later case tied at c_B goes to the human when $h(u) \leq h_B$. (Independent admission at rate r/m would leave the count binomial, free to overshoot; ranking makes the cap bind exactly.) With no positive-advantage case the policy is AI-only. Audit workload is a different object: $\hat{W}_a = n_a^{-1} \sum_i \mathbf{1}\{A_{\hat{\pi}}(X_i, U_i) = H\}$ is a realized fraction, but it is out of sample and nothing bounds it by B_0 , so the audit reports it rather than promising it.

4.5 When the audit is statistically valid

Writing T for either the assisted intervention or the frozen policy, the audit tests

$$H_0 : \Delta_H \leq 0 \text{ or } \Delta_A \leq 0, \quad H_1 : \Delta_H > 0 \text{ and } \Delta_A > 0. \quad (3)$$

Because the alternative is an intersection, the correct procedure is an intersection–union test [Berger 1982; Berger and Hsu 1996]: each margin takes an ordinary one-sided test at level $\alpha = 0.05$, both must reject, and the combined p -value is $\max\{p_H, p_A\}$. No multiplicity adjustment is needed for two co-primary margins – if either component null is true, both rejecting is contained in the true one rejecting. The same rule is the min test of Laska and Meisner [1989], analogous to component-contribution requirements in drug evaluation (21 CFR 300.50). Declaring “beats both” while a true margin is non-positive is a false audit. Adjustment does become necessary across families of budgets, policies, conditions, outcomes, or subgroups, so we pre-specify one confirmatory budget and label everything else exploratory.

PROPOSITION 4.1 (HELD-OUT AUDIT, KNOWN LOGGING PROPENSITIES). *Let $\mathcal{D}_{\text{dev}}$ and $\mathcal{D}_{\text{audit}}$ be independent and let $\hat{\pi}$, $\hat{\mu}_H$, $\hat{\mu}_A$ be functions of $\mathcal{D}_{\text{dev}}$ only. Assume consistency, conditional exchangeability ($Y^H, Y^A \perp Z \mid X$, quantitative overlap $\epsilon \leq e_z(x) \leq 1 - \epsilon$ for some $\epsilon > 0$, bounded rewards, known logging propensities, independent and identically distributed (i.i.d.) audit observations, and non-degenerate margin variances. Then, holding $\mathcal{D}_{\text{dev}}$ and n_d fixed and letting $n_a \rightarrow \infty$, $\sqrt{n_a}(\hat{\Delta}_j - \Delta_j(\hat{\pi}))_{j \in \{H, A\}} \rightsquigarrow \mathcal{N}(0, \Sigma)$, where $\rightsquigarrow$ denotes convergence in distribution and Σ is the covariance of the paired margin scores. Consequently the two component one-sided level- α tests are asymptotically valid for the particular frozen policy the development process produced, and the rule “ $L_H > 0$ and $L_A > 0$ ” has asymptotic false-audit probability at most α whenever at least one margin of the policy actually produced is non-positive.*

Conditional on $\mathcal{D}_{\text{dev}}$ the policy and regressions are fixed functions applied to an independent sample; with known propensities each score has conditional mean equal to the target value even when the frozen outcome model is wrong, so a central limit theorem applies to the paired score differences, and the unconditional statement follows by averaging over $\mathcal{D}_{\text{dev}}$. The result is asymptotic: not finite-sample exact, no license to reuse audit data for policy search, no

protection against unmeasured confounding; “audit” means a pre-specified evidentiary check, not a certification. One boundary case sits outside the non-degeneracy assumption but is conservative: an AI-only frozen policy has $d_{i,A} \equiv 0$, and the audit cannot pass.

Estimated logging propensities. If $\hat{e}_H$ is fitted on $\mathcal{D}_{\text{dev}}$ and frozen, Proposition 4.1 holds after centering at a development-dependent remainder R_n , and inference at the target margins is valid when $\sqrt{n_a} R_n = o_p(1)$; a standard sufficient condition is the product rate $\|\hat{\mu}_z - \mu_z\|_2 \|\hat{e}_z - e_z\|_2 = o_p(n_a^{-1/2})$ [Chernozhukov et al. 2018], stated in the joint regime $n_d, n_a \rightarrow \infty$ rather than the fixed- n_d regime above. Our simulation supplies the propensity analytically and reports the frozen development-estimated variant as a sensitivity check; the two agreed closely.

5 Can common choices fake “beats both”?

Yes. This section asks four questions of a simulation in which the truth is known; Table 3 maps each to its evidence and answer. Shared setup: $R = 1,000$ replications per cell of the two designs in Table 4’s caption. Scenario A is a boundary null (AI margin exactly zero, the least favorable point of H_0), so any pass there is a false audit; Scenario B sits strictly inside the null and Scenario C at a true alternative ($\Delta_A = 0.02$). All models are trained on development data alone and frozen. Monte Carlo (MC) standard error at a true rate of 0.05 is 0.0069, and every interval on a simulated rate is an MC 95% confidence interval (CI). Figure 2 draws the same rates.

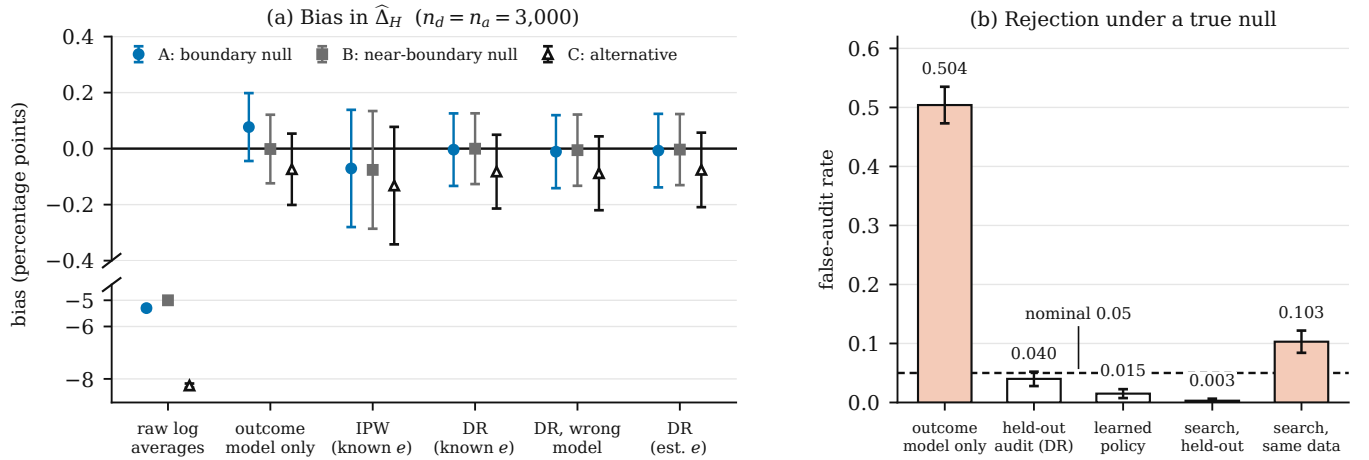
Q1. Does selective logging distort a naive comparison? Yes, by an amount comparable to the margins. The unadjusted actor-mean plug-in (freeze observed log accuracies, apply them to the candidate policy) missed Δ_H by -5.30 percentage points (pp) and Δ_A by -1.98 ; in Scenario C its error is -3.00 pp at the primary size, larger than the entire true margin of 2.0. The logging rule sends humans the cases they handle well, so the observed human baseline overstates human-only performance; an observed actor mean estimates no policy’s value. The corrections work: known-propensity IPW and DR scores recover the oracle margins within Monte Carlo error everywhere (largest $|\text{bias}|/\text{MC-SE}$ 1.7), IPW paying 1.97 $\times$ the DR sampling standard deviation, and a wrong outcome model or a frozen estimated propensity changes essentially nothing (Table 4a).

Q2. Is an unbiased estimator enough? No. The failure is in the variance. A frozen outcome-regression plug-in with audit-only uncertainty is essentially unbiased (0.00 pp on Δ_A) yet has 0.496 one-sided coverage and declared “beats both” in 0.504 of boundary-null replications (CI [0.473, 0.535]). Conditional on $\mathcal{D}_{\text{dev}}$ it is off target by the frozen model’s error, which its standard error treats as zero, so the audit reduces to the sign of that error and coverage collapses toward 1/2 rather than improving with data: 0.519 and 0.481 at $n_d = n_a = 1,000$, against 0.496 and 0.504 at triple the size. In Scenario B, strictly inside the null, the rate is still 0.406 against 0.022 for the DR audit, which with known propensities needs no repair: 0.040 (CI [0.028, 0.052]).

Q3. After a policy search, what controls the error? The held-out sample controls it. Under a complete null, selecting the best of 50 candidates and auditing on an independent $n_a = 2,000$ sample gave a false-audit rate of 0.003 (CI [0.000, 0.006]); reusing the same sample gave 0.103 (CI [0.084, 0.122]), lifting the AI-margin rejection

Table 3: What each question asks, where it is answered, and what it found. Q1–Q4 are the simulation (Section 5); Q5–Q8 the reanalysis (Section 6). Rates are false-audit rates unless stated.

Question	Evidence	Answer
Q1 Does selective logging distort a naive comparison?	Tab. 4a, Fig. 2a	Yes: the raw log-average plug-in misses Δ_H by -5.30 pp; every adjusted estimator recovers the oracle margin.
Q2 Is an unbiased estimator enough?	Tab. 4a	No: an audit-only variance rejects in 0.504 of null replications; the held-out DR audit holds at 0.040.
Q3 After a policy search, what controls the error?	Tab. 4b	The held-out sample: reuse lifts the AI-margin rejection rate from 0.059 to 0.646.
Q4 What does the standard cost?	Tab. 5	Power 0.488 against the stated alternative; audit workload exceeds the 20% budget in 0.503 of routing replications.
Q5 Do conditions beating the human also beat the AI?	Fig. 4, Tab. 6	Depends on the target: 11 of 14 beat the human baseline and 2 beat both under the broader item population (none under a familywise correction); all 14 with items fixed.
Q6 What limits resolution, participants or items?	Section 6	Items: a five-point AI margin needs ~ 118 ; participants cannot lower the item-level floor.
Q7 Does the inferential target change the verdict?	Fig. 3	It decides it: 2 of 14 against all 14, from identical point estimates.
Q8 What did the assistance cost in time?	Tab. 6	Median time moves by -2.6 to 2.9 s; the slowest condition has the smallest human margin on LSAT.

**Figure 2: Why the audit is held out. (a) Selective routing biases the raw log-average plug-in (note the axis break); every adjusted estimator recovers the oracle margin. Error bars are MC 95% intervals over $R = 1,000$ replications. (b) False-audit rates under true nulls; shaded bars mark the two procedures invalid by construction — an audit-only standard error, and auditing a searched policy on the sample that selected it.**

rate from 0.059 to 0.646 but the human margin only to 0.260. Each candidate routes a fifth of cases, so $\widehat{SE}_H$ is about $2.0 \times \widehat{SE}_A$ and the same boost moves the t -statistic half as far. Freezing the policy, not a two-margin correction, is the protection.

Q4. What does the standard cost? The cost is power, and a budget that does not transfer. Run end to end (full grid in Table 5), the protocol held the boundary-null rate to 0.015 (51% of replications were AI-only and could not pass; the routing remainder ran at 0.030), but passed only 0.488 of the time against Scenario C’s true alternative at $n_d = n_a = 3,000$ (0.164 at 1,000). Audit workload averaged 19.5% yet exceeded the 20% budget in 0.503 of routing

replications: the development cap binds exactly, but a quantile fixed on one sample is not a population cap.

6 What changes on real human-AI data?

We reanalyze the public data of Bansal et al. [2021]: 66,040 decision rows, 1,621 unique anonymized identifiers, 3 tasks (logical-reasoning questions from the Law School Admission Test, LSAT, and sentiment classification of beer and book reviews), and up to five assistance conditions per task (AI confidence alone, plus top-1, top-2, adaptive, or expert-adaptive explanations), each against the task’s human-only condition; LSAT lacks the expert arm, giving 14 pairs. This is assisted interaction, not routing, so the values

Table 4: Held-out audit behaviour under true nulls, $R = 1,000$ replications. (a) Four equally likely strata, AI accuracy 0.75, strongly selective logging ($e_H = 0.10\text{--}0.85$); a pre-specified policy routes one stratum. Scenario A sets $\Delta_A = 0$ exactly ($\Delta_H = 0.1125$); B and C sit just inside the null (-0.002) and the alternative ($+0.02$). Shown: Scenario A at $n_d = n_a = 3,000$; bias and root-mean-square error (RMSE) in percentage points; Cov. is one-sided 95% coverage of L_A ; FAR the false-audit rate. (b) Twenty identical-accuracy strata ($\Delta_H = \Delta_A = 0$); the best of 50 candidates is audited at $n_d = n_a = 2,000$. [†]Bias and RMSE only: this baseline is biased by construction.

(a) Estimating a frozen policy's value from selective logs					
	Bias $\widehat{\Delta}_H$	Bias $\widehat{\Delta}_A$	RMSE $\widehat{\Delta}_A$	Cov. L_A	FAR
Unadjusted actor mean [†]	-5.30	-1.98	2.03	—	—
Outcome regression, audit-only SE	+0.08	0.00	1.11	0.496	0.504
IPW, known e	-0.07	-0.04	2.20	0.948	0.048
DR, known e	0.00	+0.03	1.12	0.960	0.040
DR, known e , misspec. μ	-0.01	+0.03	1.11	0.957	0.043
DR, estimated e	-0.01	+0.03	1.13	0.955	0.045
(b) Searching $K = 50$ policies, then auditing the winner					
	Pr($L_A > 0$)				FAR
Held-out audit sample	0.059				0.003
Same-sample reuse	0.646				0.103

are estimated directly. Uncertainty comes from a bootstrap over participants and, for the broader target, items.

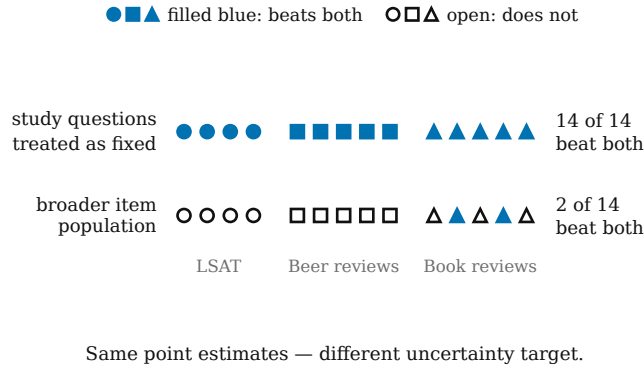


Figure 3: The main result. Markers are the 14 assisted conditions, grouped by task. Top row, fixed study questions: all 14 beat both. Bottom row, broader item population: 2 do. Same point estimates; only the uncertainty target changed. Intervals in Figure 4 (orange markers there give the fixed-item bounds); numbers in Table 6.

Q5. Do the conditions that beat the human also beat the AI? That depends on the inferential target. Every condition sits above both solo alternatives in point estimate; we reproduce those point estimates exactly and change only the uncertainty (Figure 3). With the study's questions fixed, all 14 conditions beat both baselines and survive a Holm familywise correction. Resampling items as well, the target a claim about new cases needs, 11 of 14 clear the human bound, 3 the AI bound (the narrowest by 0.02 pp), only 2 clear both,

and none survives the same correction. The correction is identical in both analyses; fixing items makes the AI value a constant, so its margin carries participant noise only. In Table 2's labels: 2 beats both, 9 beats human only, 1 beats AI only, 2 neither established.

Q6. What limits resolution — participants or items? Items, decisively. Under the broader target the AI value is a mean of per-item correctness, so its precision is bounded by the item count however many participants are recruited: the crossed bootstrap inflates the Δ_A standard error 5.1–9.2 $\times$ against a participants-only bootstrap, but Δ_H only 1.4–2.7 $\times$. With across-item standard deviations of 0.27–0.36, resolving a five-point (0.05) AI margin needs ~ 118 items (78–139) at roughly 50% power and ~ 270 at 80%, against the 20 and 50 used here. This is the stimulus-as-random-factor concern [Clark 1973; Judd et al. 2012]: define and sample the intended item population, rather than only recruiting more participants.

Q7. Does the inferential target change the verdict? It decides the verdict, and neither target is identified by the design. The verdict depends on which cases the study speaks for, its own fifty reviews and twenty questions or the broader stream they stand in for. The items were purposively assembled, not sampled (the sentiment sets were curated so that AI accuracy was 84%, comparable to human accuracy, with equal error rates, while the LSAT questions came from a prep book with their 65% AI accuracy reported as an outcome [Bansal et al. 2021]), and the participants are a crowdsourced convenience sample. We therefore report both targets and designate neither as decisive. This dependence is itself the finding. The label turns on an inferential choice that studies rarely state.

The task pattern behind the counts. On LSAT the model scores 65.0% against 58.1%. Assistance added 10.0–11.9 pp over humans, but AI margins of 3.2–5.1 pp had bounds of -9.6 to -7.6 , beats human only throughout. Book reviews are the mirror image: unaided humans (87.7%) beat the model (84.0%), and both beats-both conditions are there. The reversal lies in the participant samples, not the task. The pilots put humans above the model on all three tasks (67, 87, 85% against 65.0, 84.0, 84.0%), while the control arms came in at 58.1, 81.6, 87.7%, two of the three falling below a model accuracy fixed when the item sets were assembled [Bansal et al. 2021].

Sensitivity and data quality. We verified rather than assumed the design: every condition answered the full item pool, AI prediction and ground truth are constant within item, and all three values share item weighting. Four duplicated 50-item beer blocks (identifier collisions; 562 unique identifiers against 566 reported workers) shift no bound by more than 0.7 pp and flip no label. Assignment was randomized, but the released sample reflects post-assignment exclusions, so comparisons are conditional on retention; the pre-exclusion files give the same beats-both count (2/10/0/2 across the four labels), one knife-edge book condition trading labels. A Wald bound in place of the basic bootstrap moves 1 label, leaving the count at 2. With $B = 2,000,000$ replicates the Monte Carlo standard error of a bound is at most 0.013 pp, every bound sits at least 3 of its own Monte Carlo standard errors from zero, and the per-bound errors match the spread over eight seeds, so no label turns on the seed. Reliance measures, not co-primary: agreement 69.5–85.8%, beneficial reliance 53.2–80.6%, over-reliance 3.4–17.7%, median time 10.5–45.3 s.

Table 5: Complete simulation grid, $R = 1,000$ replications per cell. (a) Pre-specified-policy audit: unadjusted bias on Δ_A (pp), pass rates per estimator (OutReg is the audit-only-variance plug-in of Table 4), and one-sided 95% coverage for the known-propensity DR score. (b) End-to-end learned policy at the 20% cap: pass rate, mean audit workload, AI-only share, and the share of all replications exceeding the cap. Column n gives $n_d = n_a$. Under A and B a pass is a false audit; under C it is power.

Scenario	n	(a) pre-specified policy						(b) learned policy, DR known e			
		$\text{Bias}_{\Delta_A}^{\text{unadj}}$	OutReg	IPW	DR	$\text{DR}_{\text{est } e}$	Cov. DR	pass	$\widehat{W}_a$	AI-only	>cap
A (boundary null)	1,000	-1.96	0.481	0.044	0.025	0.026	0.972	0.025	10.8%	46%	0.271
	3,000	-1.98	0.504	0.048	0.040	0.045	0.960	0.015	9.8%	51%	0.245
B (near null)	1,000	-1.89	0.453	0.037	0.030	0.034	0.959	0.012	10.5%	48%	0.255
	3,000	-1.88	0.406	0.044	0.022	0.018	0.966	0.016	8.2%	59%	0.191
C (alternative)	1,000	-3.06	0.853	0.089	0.222	0.219	0.961	0.164	17.5%	13%	0.447
	3,000	-3.00	0.969	0.228	0.581	0.567	0.952	0.488	19.5%	3%	0.490

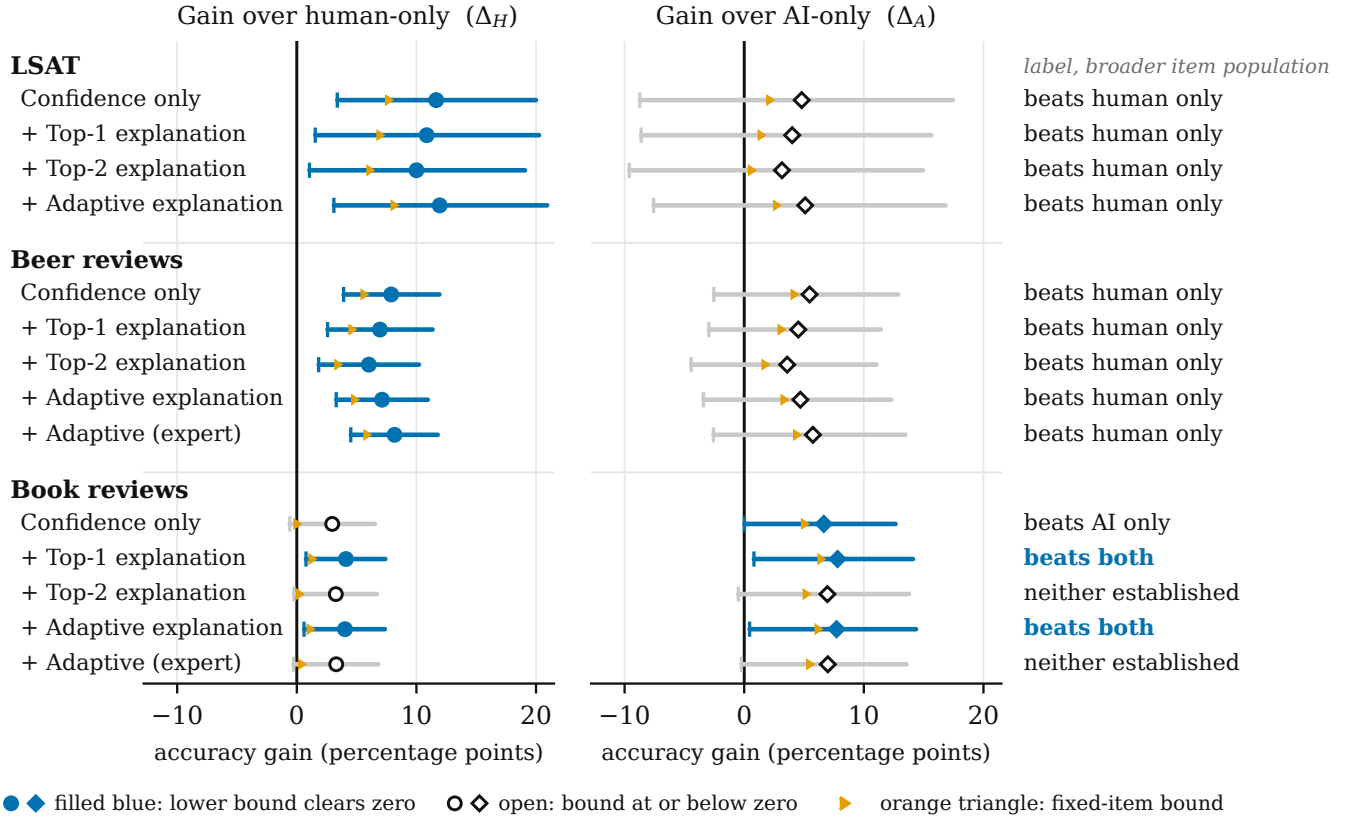


Figure 4: Intervals behind Figure 3. Rows are accuracy gains over human-only (left) and AI-only (right) on common items; bars are 90% basic-bootstrap intervals whose lower endpoints are the one-sided 95% bounds. Under the fixed-item bounds (orange) all 14 conditions clear both margins; none survives a familywise correction under the broader target.

Q8. What did the assistance cost in human time? Median response times move by -2.6 to 2.9 s against the same task’s human-only condition. The top-2 explanation condition is the slowest on both LSAT and book reviews, with the smallest human margin on LSAT and tied second-smallest on book reviews. Beer’s expert-adaptive condition shows the largest saving (-2.6 s) and the largest human margin in its task. A performance table without this column omits half the trade-off.

7 Discussion and limitations

What the audit changes. TEAM-Audit changes the default reporting standard, not the design space: both solo comparisons, uncertainty, held-out evaluation after any search, a selective-log correction, workload beside accuracy, and labels that match what was shown. Every ingredient is standard; the contribution is the protocol and the vocabulary it licenses.

Table 6: Two-margin results for every assisted condition of Bansal et al. [2021] (percentage points, common items; margins use unrounded values). L_H and L_A are one-sided 95% basic-bootstrap lower bounds ($B = 2,000,000$ crossed replicates); the audit passes only when both exceed zero. Δt is median time minus the human-only median. Labels follow Table 2 (bold = beats both) and are descriptive: a Holm correction retains none of the 14 under this target and all 14 with items fixed. A beats-human-only label means the AI margin was unresolved at this item count, not that complementarity was ruled out.

Task	Condition	Participants		Items	Team	Human	AI	$\hat{\Delta}_H$	L_H	$\hat{\Delta}_A$	L_A	Δt	Label
		team	human										
(s)													
LSAT	Confidence only	102	100	20	69.8	58.1	65.0	+11.7	+3.38	+4.8	-8.73	-2.2	beats human only
	+ Top-1 expl.	101	100	20	69.0	58.1	65.0	+10.9	+1.54	+4.0	-8.61	+0.3	beats human only
	+ Top-2 expl.	103	100	20	68.2	58.1	65.0	+10.0	+1.06	+3.2	-9.61	+2.9	beats human only
	+ Adaptive expl.	101	100	20	70.1	58.1	65.0	+11.9	+3.10	+5.1	-7.57	-0.1	beats human only
Beer reviews	Confidence only	92	92	50	89.5	81.6	84.0	+7.9	+3.92	+5.5	-2.54	-0.5	beats human only
	+ Top-1 expl.	88	92	50	88.5	81.6	84.0	+7.0	+2.57	+4.5	-2.95	-1.6	beats human only
	+ Top-2 expl.	92	92	50	87.6	81.6	84.0	+6.0	+1.83	+3.6	-4.43	-0.9	beats human only
	+ Adaptive expl.	98	92	50	88.7	81.6	84.0	+7.1	+3.30	+4.7	-3.41	-0.6	beats human only
	+ Adaptive (expert)	100	92	50	89.7	81.6	84.0	+8.2	+4.51	+5.7	-2.57	-2.6	beats human only
Book reviews	Confidence only	92	91	50	90.7	87.7	84.0	+3.0	-0.59	+6.7	+0.02	+0.5	beats AI only
	+ Top-1 expl.	91	91	50	91.8	87.7	84.0	+4.1	+0.77	+7.8	+0.81	+1.1	beats both
	+ Top-2 expl.	90	91	50	91.0	87.7	84.0	+3.3	-0.24	+7.0	-0.49	+2.3	neither established
	+ Adaptive expl.	93	91	50	91.7	87.7	84.0	+4.0	+0.60	+7.7	+0.45	0.0	beats both
	+ Adaptive (expert)	95	91	50	91.0	87.7	84.0	+3.3	-0.28	+7.0	-0.23	+0.8	neither established

What the audit does not claim. It does not define teamwork: shared goals, communication, adaptation, agency, accountability, trust, and fairness are outside its scope [Klein et al. 2004; National Academies of Sciences, Engineering, and Medicine 2022; O’Neill et al. 2022], and routing can beat both baselines without establishing any of them. The narrow scope is deliberate. If the teaming construct is doing work in a performance argument, that work is testable, and failing means the claim was not established, not disproved. Two consequences follow. Beating AI-only presumes full automation is available; where it is not [De-Arteaga et al. 2020; Green and Chen 2019], Δ_A is a benchmark rather than a decision. And a standard this strict (ours passes under half the time against a true alternative) will be invoked selectively, so who runs the audit, and against whose claim, is part of what the construct does.

Limitations. Performance is a narrow lens and accuracy a narrow outcome. Selective-log inference rests on consistency, overlap, and control of measured confounders, none testable from the log alone. The audit covers a frozen policy, not an adapting router, and Proposition 4.1 is asymptotic, conditional, and pointwise, so a margin with merely near-zero variance can still need a large audit sample before normality binds. The 20% cap binds on development only; audit workload exceeds it in 0.503 of routing replications. The simulation validates the delegation estimators only; the crossed bootstrap is standard but not separately simulated, and its item component rests on as few as 20 items. The assisted family is descriptive (confirmatory use would require pre-specifying the condition), and the displayed contrasts are conditional on retention rather than intention-to-treat estimands. We audit no deployed router, because logged propensities are rarely recorded, which is itself a finding. All estimates are specific to this sample, item set, AI version, outcome, and design, and terminology remains normative: passing a performance audit does not settle whether “team” is the right metaphor.

8 Conclusion

Human participation is not itself evidence of “better together.” The performance version of that claim has an exact form — the combined system beats the human alone and beats the AI alone — and TEAM-Audit tests exactly that claim: freeze the system, evaluate it held out, correct selectively logged outcomes, and require both one-sided bounds to clear zero, with human workload reported beside accuracy. In simulation, reused data or incomplete uncertainty manufactures “beats both” under a true null; on real data the verdict turns on which cases the claim is about (14 of 14 against 2). The audit does not define social teamwork; it makes one implication of the metaphor testable, and shows how much disagreement about the metaphor is really disagreement about which population the evidence describes.

Reproducibility. Code and locked result files regenerating every number in this paper: <https://github.com/HamedKhosravi99/TEAM-Audit>.

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